

434/868/915 MHz SoC Wireless Transparent Transmission Module

Product Overview

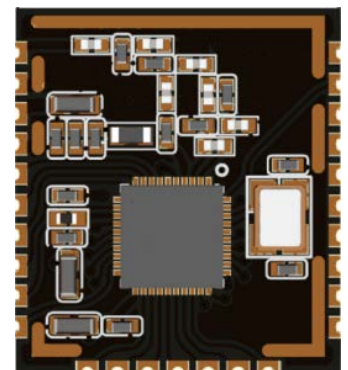
HM-380F64 is a wireless data transparent transmission module based on the CMT2380F64 ultra-low power sub-1GHz wireless transceiver SoC chip. It has the advantages of ultra-low power consumption, high sensitivity, long-distance communication and high performance. Moreover, the module has the transparent transmission function, users can quickly realize wireless data transmission without taking time to get the complex radio frequency configuration. The module integrates rich peripheral functions and supports standard UART, I2C and SPI interfaces, RTC, timer, DMA and 12-bit high-speed ADC, etc. Supports multiple wireless data packet formats and encoding / decoding methods, up to 64-byte Tx/Rx FIFO, rich-functional radio frequency GPIO, multiple low-power operation modes and quick start mechanism, high-precision RSSI, and manual fast frequency hopping to meet needs of multi-directional secondary development by users. The product is divided into three operating frequencies: 434MHz, 868MHz, and 915Mhz, which supports data rate range of 0.5-300kbps.

PRODUCT FEATURES

- Super anti-interference ability, suitable for use in complex interference environments
- Receive sensitivity: -120dBm @ 434MHz, 0.6 kbps, GFSK
- Operating frequency: 434MHz, 868MHz, 915MHz
- Power supply voltage input range: 1.8 V-3.6 V
- Transmitting current: 74 mA @ + 20 dBm, 434 MHz
- Received current: 12mA @ 434MHz, FSK
- Deep sleep current: $\leq 1.5 \mu\text{A}$
- Operating rate: 0.5-300 kbps

Application

- Automatic meter reading
- Furniture security and building automation
- Wireless Sensor Node and Industrial Monitoring
- ISM frequency band data communication



HM-380F64

Ordering information

Part Number	Operating frequency
HM-380F64-433S2	434MHz
HM-380F64-868S2	868MHz
HM-380F64-915S2	915MHz

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1. Module Pin Diagram

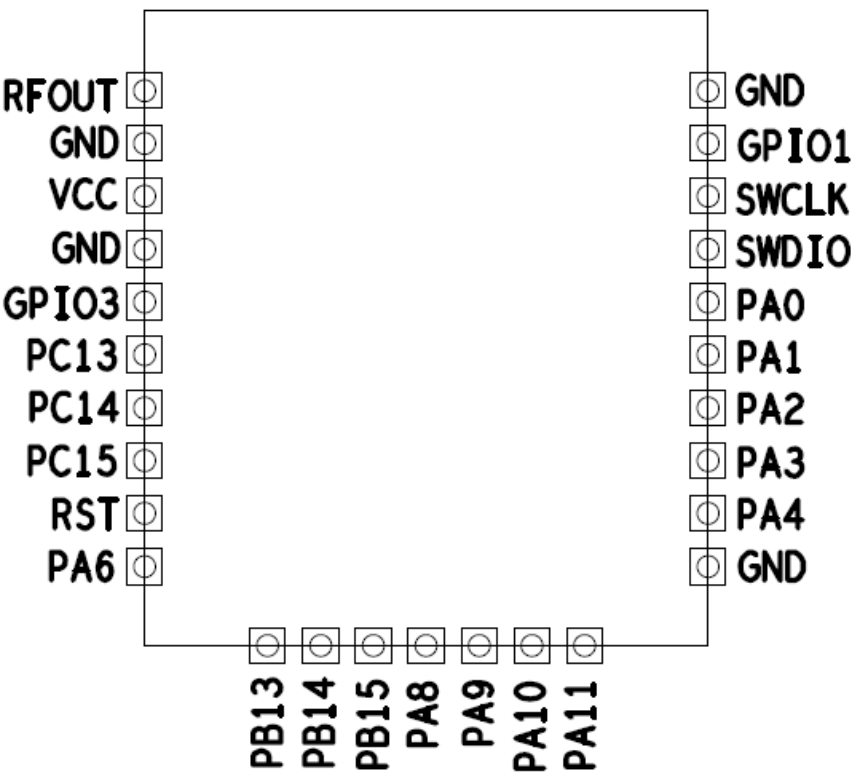


Figure 1. HM-380F64 module front view

2. Pin Definition

Pin Description	Name	Pin Characteristics	Function Description
1	RFOUT	--	RF output
2	GND	--	Power supply negative /ground
3	VCC	--	1.8 V-3. 6V power supply positive
4	GND	--	Power supply negative /ground

5	GPIO3	O	Mapping DOUT output in receive mode
6	PC13	IO	RTC _ TAMP1, RTC _ TS, RTC _ OUT, WKUP1
7	PC14	IO	OSC32_IN
8	PC15	IO	OSC32_OUT
9	RST	I	Hardware reset pin, low active
10	PA6	IO	SPI1 _ MISO, TIM3 _ CH1, TIM1 _ BKIN, TIM8 _ CH1, EVENT _ OUT, LPUART _ CTS, LPUART _ TX, I2C2 _ SCL, LPTIM _ ETR, BEEPER _ OUT, COMP _ OUT, ADC _ IN6, OPAMP _ VOUT
11	PB13 ^[1]	I	Module working mode selection port 0
12	PB14 ^[2]	I	Module working mode selection port 1
13	PB15	IO	SPI _ MOSI, SPI 2 _ MOSI, I2S _ SD, TIM1 _ CH3N, TIM8 _ CH3N, TIM8 _ CH4, RTC _ REFIN
14	PA8	IO	USART1 _ CK, TIM1 _ CH1, EVENT _ OUT, MCO, SPI2 _ NSS, TIM8 _ CH2N
15	PA9	O	Serial port TXD
16	PA10	I	Serial port RXD
17	PA11	I	USART1 _ CTS, TIM1 _ CH4, EVENT _ OUT, I2C2 _ SCL, SPI2 _ MOSI, COMP _ OUT
18	GND	--	Power supply negative /ground
19	PA4	O	Normal output high when module is in running and wake-up mode, low when entering configuration mode/low when module is in low power consumption and sleep mode
20	PA3	IO	USART1 _ RX, USART2 _ RX, TIM8 _ CH4, TIM1 _ CH2, SPI1 _ MISO, I2S _ MCLK, LPUART _ RX, COMP _ INP, ADC _ IN3
21	PA2	I	Module in Run and Wakeup Mode pull down PA2 to enter into Module Configuration/ Module in low power and sleep mode pull down PA2 to provide a negative pulse signal to wake up module.
22	PA1	IO	USART1 _ RTS, EVENT _ OUT, SPI1 _ NSS, I2S _ WS, I2C1 _ SMBA, LPTIM _ IN2, LPUART _ TX, TIM8 _ CH2, TIM3 _ ETR, COMP _ INP, ADC _ IN1, OPAMP _ VINP
23	PA0 ^[3]	IO	USART1 _ CTS, USART2 _ CTS, USART2 _ RX, LPUART _ TX, LPUART _ RX, SPI1 _ SCK, I2S _ CLK, LPTIM _ IN1, TIM8 _ CH1, RTC _ TAMP2, WKUP0, COMP _ INM, COMP _ OUT, ADC _ IN0, OPAMP _ VINP
24	SWDIO	SW Debug Data Interface	PA13, USART1 _ TX, USART1 _ RX, USART2 _ RX, I2C1 _ SDA, SPI1 _ SCK, I2S _ CLK

25	SWCLK	SW Debug Data Interface	PA14, USART1 _ TX, USART2 _ TX, I2C1 _ SMBA, SPI1 _ MISO
26	GPIO1	GPIO1 of RF	Mapping function can be configured by AT command
27	GND	--	Power supply negative /ground

Note:

[1] Level combination of PB13 and PB14 refers to working mode configuration.

[2] Level combination of PB13 and PB14 refers to working mode configuration.

[3] The module comes with production and testing firmware from the factory. It will pull down PA0 externally to enter into production test and then close the debugging interface. If the second development of the module is required, please be careful not to lower PA0 externally when burning at the first time to avoid burning failure.

Table 1. HM-380F64 module pin definition

Note:The module SPI interface test point is interconnected inside the chip. Generally, it is not used by default. The pin definition is shown in the figure below.

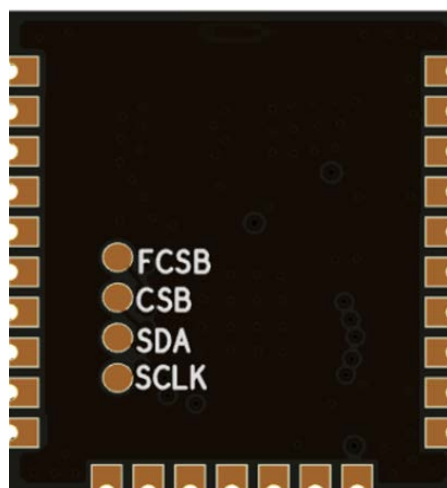


Figure 2. HM-380F64 Module Test Point Definition

3. Module application connection diagram

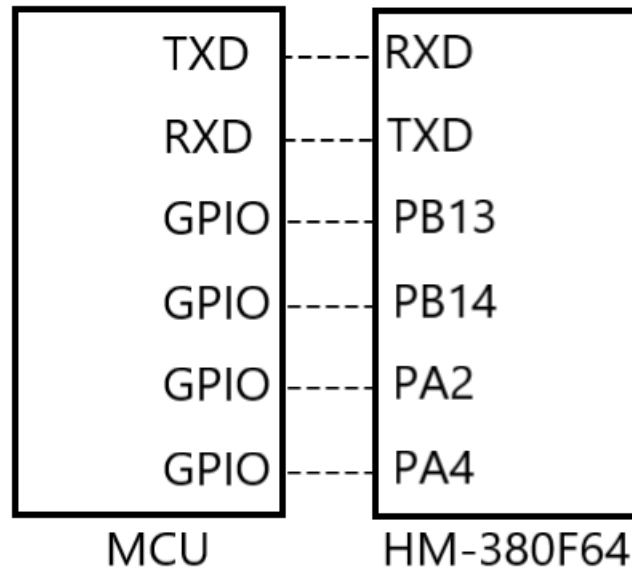


Figure 3. Module application connection diagram

The basic application of the module is generally connected to six pins. TXD and RXD are the communication pins of serial port data, PB13/PB14 are used to switch the module working mode, PA2 is used to wake up after the module enters the configuration mode and the sleep mode, and PA4 is used to output the module status signal.

If there is no need to switch the working mode while using, user can pull the PB13/PB14 pin externally to the corresponding working mode level without entering into the configuration mode and module status signal indication, and PA2/PA4 can be directly suspended. So as to save the IO resources of the upper computer MCU.

4. Working Mode Configuration

The module has four working modes, which can be selected by different level combinations of PB13 and PB14:

PB14	PB13	Status	Function Description
1	1	Mode of operation	In this mode, the MCU and RF will always be in the working state. Under normal conditions, the RF will always be in the receiving state and send the data from the serial port immediately after receiving it. If the data is received from the serial port, the RF will switch to the transmitting state and transmit the data.
0	1	Wake-up mode	This mode is similar to the operating mode, except that there will be a long preamble to wake

			up the module in low power mode while transmitting, thus a long time will be taken to transmitting a packet of data, which depends on the low power period set by the user. This mode is mainly used to communicate with modules in low power mode.
1	0	Low power mode	In this mode, the MCU will enter into sleep state, that is, the serial port of the module cannot receive data. RF will be in a cyclic working state of sleep and receiving, and the time of the low power consumption period is set by users. The received data in this mode will output a 5ms high level from the PA4 pin and then output data from the serial port.
0	0	Sleep Mode	In this mode, both the MCU and RF enter into sleep state, and the power consumption reaches the lowest.

Table 2. HM-380F64 Working Mode Configuration

The four modes can be switched with each other, but the low power mode and sleep mode cannot immediately respond to mode changes. You need to set the level of the mode pins (PB13, PB14) first and then give a negative pulse to the wake-up pin (PA2) to wake up the module and switch to the corresponding working mode.

The module status indicates the PA4 pin to stay high in working mode and wakeup mode, and remains low in low power mode and sleep mode. User can get to know whether the module mode has been switched by detecting the pin status.

In addition, it should be noted that when switching from running mode to sleep mode, you should first switch to wake-up mode and then switch to sleep mode. Because when switching to low power mode first, the module will enter into sleep state and cannot respond to subsequent mode level signals.

5. Electrical parameters

Test conditions: Power supply is 3.3 V, operating temperature is 25 °C

Parameter	Symbols	Status	Minimum	Typical	Maximum	Units
Operating frequency	F _C	HM-380F64-433S2		434		MHz
		HM-380F64-868S2		868		MHz
		HM-380F64-915S2		915		MHz
Modulation mode	OD		GFSK			

Receive sensitivity	ENS	434 MHz, DR = 2.0 kbps, $F_{DEV}= 10$ KHz		118		B m
		868 MHz, DR = 2.0 kbps, $F_{DEV}= 10$ KHz		116		B m
		915 MHz, DR = 2.0 kbps, $F_{DEV}= 10$ KHz		115		B m
Data rate	R		5	4	00	b p s
Receive bandwidth	W		0		30	H z
Operating voltage	DD		8	3	6	
Receive operating current	Rx	434MHz		0	5	A
		868MHz		0	5	A
		915MHz		0	5	A
Receive operating current	Tx	434MHz + 20dbm		3	5	A
		868MHz + 20dbm		1	5	A
		915MHz + 20dbm		1	5	A
sleep current	Sleep			4		A
Image frequency immunity	MR	$F_{RF}= 433$ MHz		5		Bc
		$F_{RF}= 868$ MHz		3		Bc
		$F_{RF}= 868$ MHz		3		Bc
Working temperature	OP		40		85	

Table 3. Electrical parameters

6. Module outline dimension

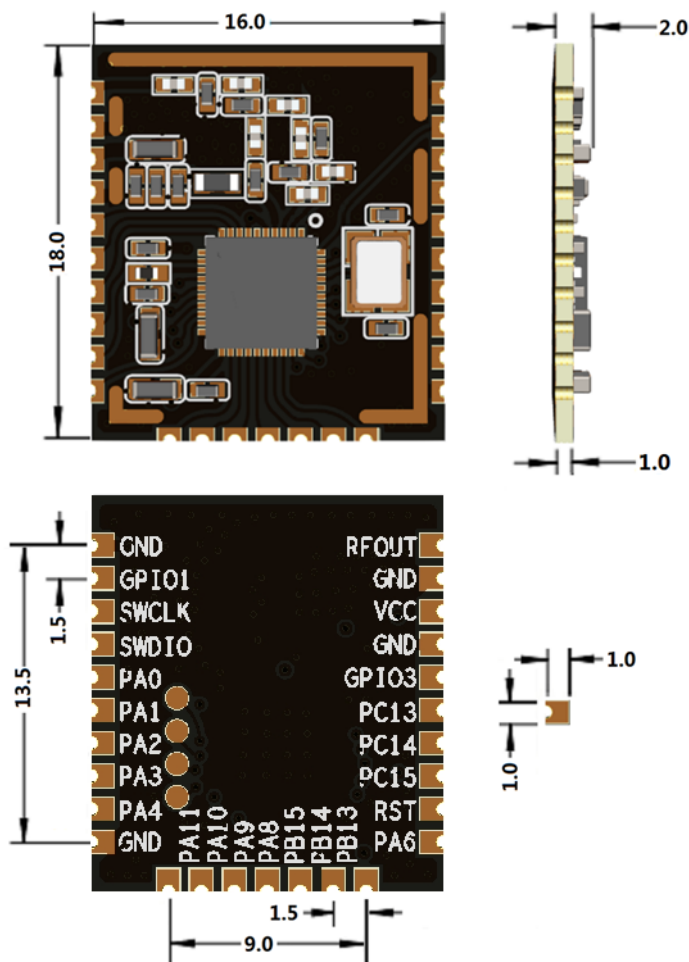


Figure 4. Module scale (Unit: mm)

7. Revise Record

Document Version	Update Date	Update Content
V1.0	2024. 4. 17	Initial

8. Contacts

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